

Factory Pattern Sample Code

1-Create interface

```
ExportFile.java X
factory-pattern-springboot > src/main/java > com.pisethjavaschool.factory_pattern_springboot.export_homework11 > ExportFile >
1 package com.pisethjavaschool.factory_pattern_springboot.export_homework11;
2
3 public interface ExportFile {
4     void export(String content);
5 }
6
```

2-Create component

```
ExportFile.java  CsvExportFile.java X
factory-pattern-springboot > src/main/java > com.pisethjavaschool.factory_pattern_springboot.export_homework11 > CsvExportFile > export(String) : void
1 package com.pisethjavaschool.factory_pattern_springboot.export_homework11;
2
3 import org.springframework.stereotype.Component;
4
5 @Component("CSV")
6 public class CsvExportFile implements ExportFile {
7
8     @Override
9     public void export(String content) {
10         System.out.println("Export CSV: " + content);
11     }
12 }
13
14 }
15
```

យើងគ្រាន់តែបង្កើត Component ហើយដាក់ឈ្មោះជាទៅចឹងវានឹងបង្កើត bean name យើងហើយ

```
1 package com.pisethjavaschool.factory_pattern_springboot.export_homework11;
2
3 import org.springframework.stereotype.Component;
4
5 @Component("EXCEL")
6 public class ExcelExportFile implements ExportFile {
7
8     @Override
9     public void export(String content) {
10         System.out.println("Export Excel: " + content);
11     }
12 }
13
14 }
```

```

package com.pisethjavaschool.factory_pattern_springboot.export_homework11;

import org.springframework.stereotype.Component;

@Component("PDF")
public class PdfExportFile implements ExportFile {

    @Override
    public void export(String content) {
        System.out.println("Export PDF: " + content);
    }

}

```

```

package com.pisethjavaschool.factory_pattern_springboot.export_homework11;

import org.springframework.stereotype.Component;

@Component("JSON")
public class JsonExportFile implements ExportFile {

    @Override
    public void export(String content) {
        System.out.println("Export JSON: " + content);
    }

}

```

3- Create Factory កន្លែងដែលបង្កើត Object អោយយើង

```

package com.pisethjavaschool.factory_pattern_springboot.export_homework11;

import java.util.Map;

@Component
@RequiredArgsConstructor
public class ExportFileFactory {

    private final Map<String, ExportFile> exportFiles;

    public ExportFile getExportFile(String type) {
        ExportFile exportFile = exportFiles.get(type);

        if (exportFile == null) {
            throw new IllegalArgumentException("Unsupported export type: " + type);
        }

        return exportFile;
    }

}

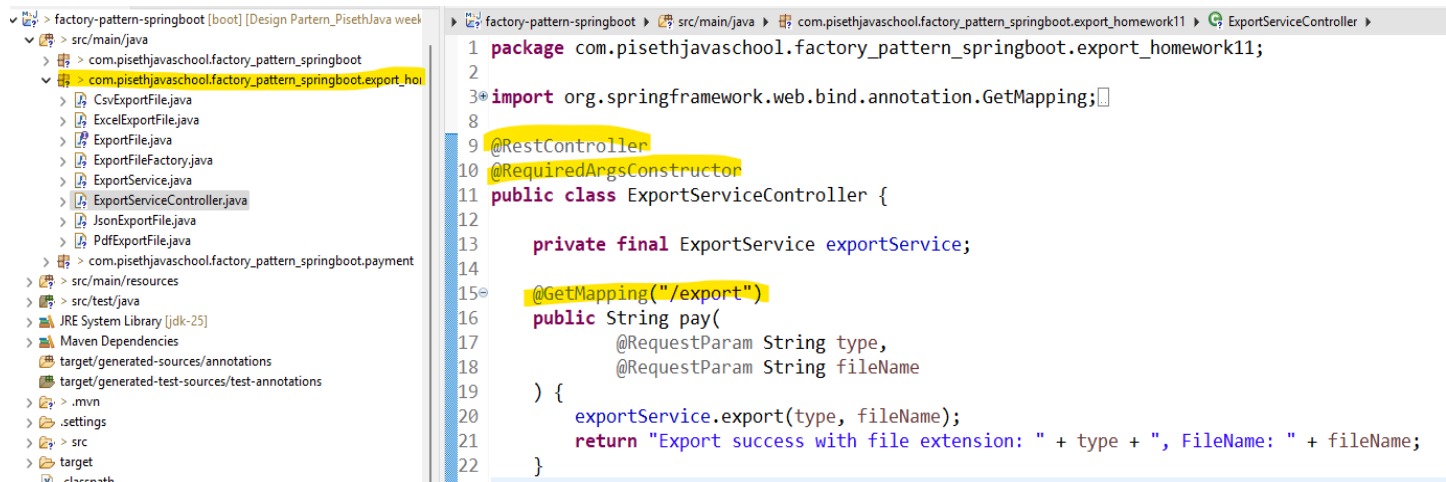
```

កន្លែង Factory នឹងគេវាបានប្រើប្រាស់ Map ដើម្បីធ្វើការ អោយវាដាក់ String and value វាចូលដើម្បីបង្កើត Object

4- Create Service

```
1 package com.pisethjaschool.factory_pattern_springboot.export_homework11;
2
3 import org.springframework.stereotype.Service;
4
5
6
7 @Service
8 @RequiredArgsConstructor
9 public class ExportService {
10
11     private final ExportFileFactory factory;
12
13     public void export(String type, String content) {
14
15         ExportFile exportFile = factory.getExportFile(type);
16
17         exportFile.export(content);
18     }
19 }
20
21
```

5- Controller



factory-pattern-springboot [boot] [Design Pattern, PisethJava week]

- > src/main/java
 - > com.pisethjaschool.factory_pattern_springboot
 - > com.pisethjaschool.factory_pattern_springboot.export_homework11
 - > CsvExportFile.java
 - > ExcelExportFile.java
 - > ExportFile.java
 - > ExportFileFactory.java
 - > ExportService.java
 - > ExportServiceController.java
 - > JsonExportFile.java
 - > PdfExportFile.java
 - > com.pisethjaschool.factory_pattern_springboot.payment
 - > src/main/resources
 - > src/test/java
 - > JRE System Library [jdk-25]
 - > Maven Dependencies
 - > target/generated-sources/annotations
 - > target/generated-test-sources/test-annotations
 - > .mvn
 - > .settings
 - > > src
 - > target
 - > .classpath

```
1 package com.pisethjaschool.factory_pattern_springboot.export_homework11;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4
5
6
7 @RestController
8 @RequiredArgsConstructor
9 public class ExportServiceController {
10
11     private final ExportService exportService;
12
13     @GetMapping("/export")
14     public String pay(
15         @RequestParam String type,
16         @RequestParam String fileName
17     ) {
18         exportService.export(type, fileName);
19         return "Export success with file extension: " + type + ", FileName: " + fileName;
20     }
21 }
22
```